Codrin Crismariu

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Education

Tufts University Expected 2028
Mechanical Engineering GPA: 3.97/4

Research Experience

Research Intern February 2026 – Present
SPARC Lab, Department of Mechanical Engineering, Tufts University *Medford, MA*

- Led the end-to-end development of MARCH and first-authored the resulting CoRL 2026 paper, bridging model-based foothold planning and reinforcement learning to enable vision-guided humanoid locomotion.
- Built a MuJoCo/mjlab sim-to-real pipeline with 4,096 parallel environments, domain randomization, PPO teacher training, and CNN–transformer–MDN policy distillation; achieved 66.7% success across 1,024 randomized simulation trials while halving teacher-training episodes versus a model-free baseline.
- Deployed the policy at 50 Hz on a Unitree G1's Jetson Orin NX using RealSense depth and proprioception, demonstrating traversal of four laterally constrained stepping stones while reducing joint torque by at least 12% and angular jerk by 39%.
- Currently developing a general-purpose model that learns transferable tactile latent representations from images and video for manipulation-uncertainty detection and augmentation of vision-language-action models.

Publications

Codrin Crismariu and Ryan K. Cosner. "MARCH: Model-Assisted Reinforcement Learning for the Perceptive Control of Humanoids over Sparse Footholds." Accepted to the *Conference on Robot Learning (CoRL)*, 2026. [arXiv]

Industry Experience

Advanced Robotics R&D Intern Summer 2026
Magna International *Troy, MI*

- Evaluated the use of humanoid robots for payload manipulation and material transport, identifying requirements and deployment opportunities in factory environments.
- Developed a pipeline for fine-tuning and deploying vision-language-action models on the Unitree G1 humanoid platform for dexterous industrial tasks.
- Paired VLA policies with diffusion-steering reinforcement learning to create a self-improving agent designed for deployment in real manufacturing environments.

Technical Skills

Programming & Frameworks: Python, C++, MuJoCo, mjlab, Isaac Lab, rsl-rl, LaTeX